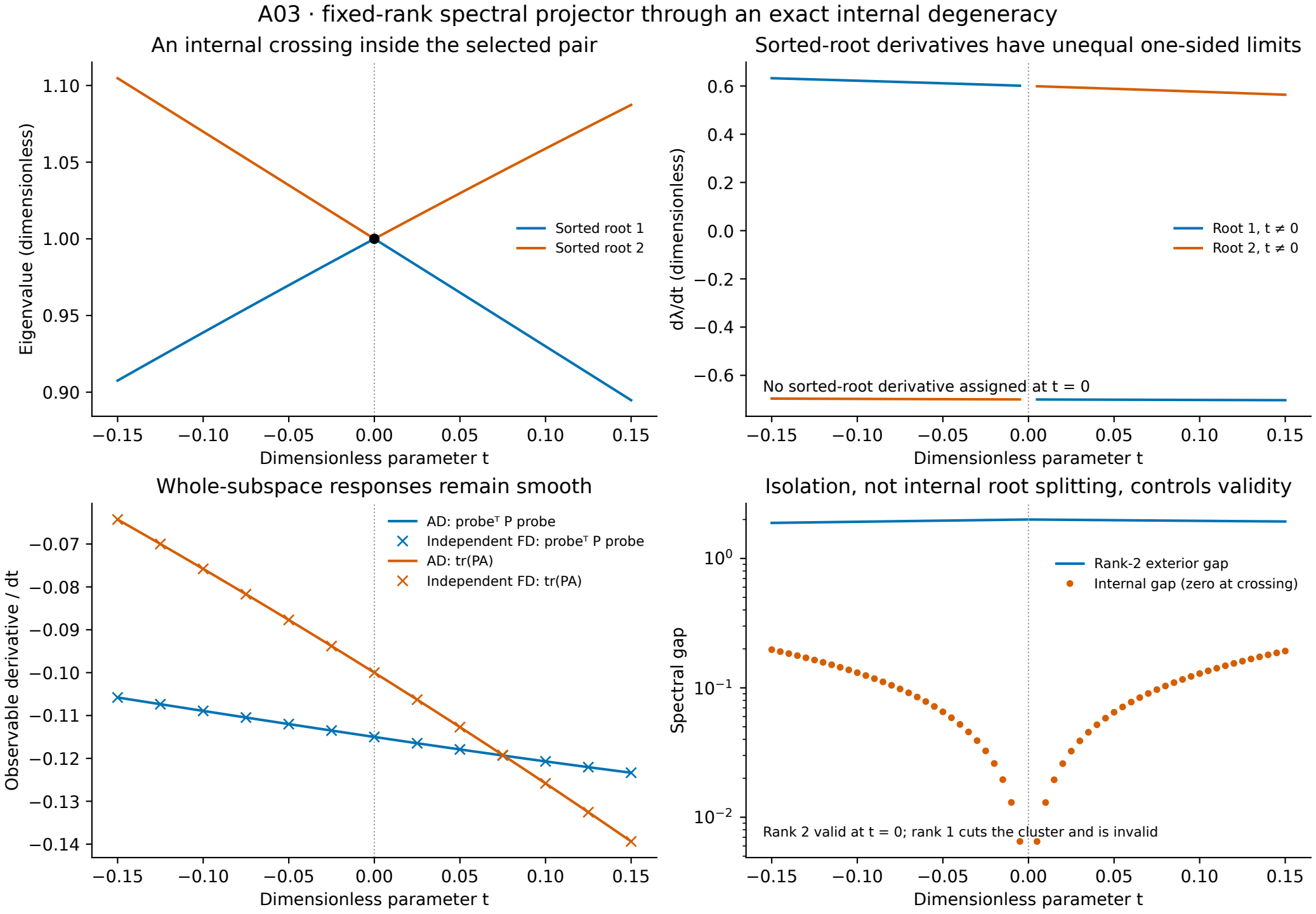
A03 · fixed-rank spectral projector through an exact internal degeneracy
An internal crossing inside the selected pair
Eigenvalue (dimensionless)
Dimensionless parameter t
Sorted root 1
Sorted root 2
Sorted-root derivatives have unequal one-sided limits
dλ/dt (dimensionless)
Root 1, t ≠ 0
Root 2, t ≠ 0
No sorted-root derivative assigned at t = 0
Whole-subspace responses remain smooth
Observable derivative / dt
AD: probeᵀ P probe
Independent FD: probeᵀ P probe
AD: tr(PA)
Independent FD: tr(PA)
Isolation, not internal root splitting, controls validity
Spectral gap
Rank-2 exterior gap
Internal gap (zero at crossing)
Rank 2 valid at t = 0; rank 1 cuts the cluster and is invalid